REVIEW

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Deep learning for crowd counting in complex environments: challenges and novel trends

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Abstract
Crowd counting is a challenging task, especially when accurately estimating the number of people in large crowds under diverse environmental conditions. Factors such as harsh lighting, crowded spaces, adverse weather, and varying perspectives make it difficult for traditional methods to count individuals reliably. Modern solutions leverage deep learning models, including hybrid approaches, to overcome these challenges. A significant advancement in this area is the use of attention mechanisms, which enable models to focus on the most relevant regions of the crowd, improving counting accuracy. Additionally, curriculum learning, which gradually introduces complexity during training, enhances model performance in unpredictable environments. Another notable improvement is the combination of Generative Adversarial Networks with the U-Net architecture, which generates synthetic data to improve training and generalization. Hybrid deep learning approaches that integrate adaptive curriculum learning and attention mechanisms have shown promising results in handling diverse scenarios. Moreover, incorporating fuzzy methods into preprocessing enables better handling of varying densities, crowd behaviors, and illumination conditions, leading to more accurate crowd counting across a wide range of environments. By integrating these advanced techniques, researchers can address current limitations and improve the robustness and real-world applicability of crowd counting models.

Keywords Crowd counting, Deep learning (DL), Dense crowd monitoring, Detection-based tracking, Lightweight network, Crowd density estimation

1 Introduction

A good way to define a crowd is by distinguishing it from a group, as illustrated in Table 1. A group is a collection of people, ranging from two to hundreds, who are together at the same time, interacting socially. They stay close to each other, move at similar speeds, and in the same direction. A crowd denotes a large group of people who have gathered in one spot, often sharing a common purpose or interest. Individual identities can merge into a collective entity, leading to shared behaviors and actions, which are discussed in [1].

Crowd analysis involves estimating the number and spatial distribution of people in areas such as social places. CC becomes challenging in harsh scenes; these issues require

Table 1 Group and crowd characteristics: a comparative overview

Aspect	Group	Crowd
Purpose	Formed with a specific purpose or goal	Often forms spontaneously or for an event/situation
Size and structure	Generally smaller and more manageable	Larger in size and often lacks formal structure
Identity and unity	Members maintain individual identities while contributing to group identity	Individual identities may merge into a collective identity
Duration and stability	More stable over time with ongoing membership	Based on condition changes
Examples	Work teams, clubs, and organizations	Sporting and social occasions

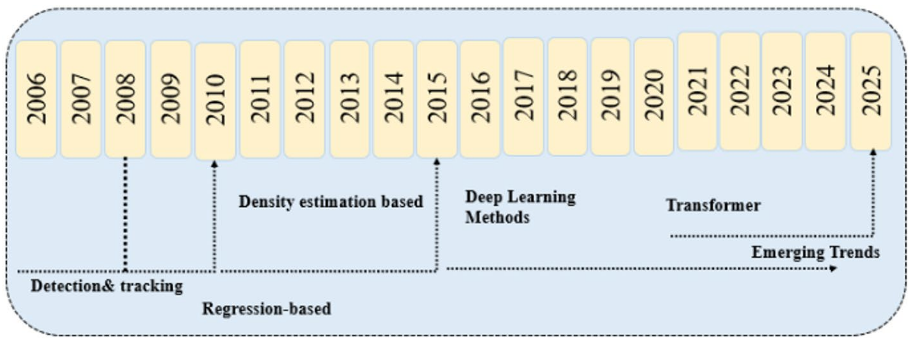


Fig. 1 A concise timeline of crowd counting developments

advanced DL methods for accurate results. Crowd density estimation (CDE) focuses on generating density maps to show people's distribution, which are useful for traffic and safety management, which are introduced in [2]. CNN-based methods, which are presented in [3] are widely used for generating density maps, while transformer-based models are proposed in [4]. GAN-UNet utilizes GANs to generate realistic crowd images and enhance density maps in unevenly distributed areas. Various training strategies enhance CC model performance: Curriculum Learning improves generalization by gradually increasing task difficulty; Standard DL is simple but may struggle with complex cases; Adaptive Learning adjusts to data complexity but requires monitoring; and Multi-task Learning improves accuracy by training on related tasks like movement or density, but adds complexity [5].

A chronology of CC methods is shown in Fig. 1. Traditional approaches (pre-2015) included regression-based techniques such as Multi-Level Regression (MLR), Kernel Ridge Regression (KRR), and Ridge Regression (RR), that shown in [6]. Chan et al. applied learning-based estimation, while Lempitsky et al. reviewed in [7] combined regression with feature learning. Context-Aware Ridge Regression (CA-RR) and Count are discussed in [8] improved performance by using context and decision forests. Cross-scene counting was explored by Wang et al. in [9] and Fu et al. presented [10]. In 2015, DL methods emerged with MCNN and Hydra-CNN [11], which introduced multi-branch architectures. Between 2016 and 2018, models like Switching CNN, CP-CNN, and CSRNet were introduced in [12], and SANet advanced CC through mechanisms like switching, contextual pyramids, and dilated convolutions. Recent models (2019–present) include PSSDN, which was published in [13], which uses point-level supervision. Transformer-based models analyzed in [14], which have become promising alternatives to CNNs due to their self-attention mechanism. TransCrowd, that discussed in [15] uses weak supervision, CrowdFormer combines Transformers with overlapping convolutions, and other hybrid models, which are covered in [16] merge Transformer and CNN features or use local attention to better handle dense crowds and complex backgrounds.

In preparing this review, we adopted a narrative review approach while ensuring a structured and transparent selection process. Relevant articles were identified through searches in major scientific databases, including IEEE Xplore, Scopus, Web of Science, and Google Scholar. The search covered publications from 2010 to 2025 using keywords such as crowd counting, density estimation, computer vision, deep learning, and surveillance. To maintain quality, we included only peer-reviewed journal articles and reputable conference proceedings, while excluding preprints and non-scholarly sources. Furthermore, studies were selected if they directly addressed crowd counting methods, datasets, or applications, whereas works with limited relevance or lacking methodological contributions were not considered. Although this work is narrative rather than systematic, we strived to provide comprehensive and balanced coverage of the most influential and recent contributions in the field.

1.1 This work offers the following major contributions

In this review, we present various crowd-counting methods, including detection-based approaches for sparse crowds, regression-based methods for density mapping, and density-estimation techniques for complex scenarios. We highlight recent advancements, including conventional neural networks for feature extraction, attention mechanisms for improved focus, and multi-scale approaches to handle density variations. Detection and tracking techniques are also emphasized, enabling accurate localization and monitoring of individuals in videos, which is essential for real-time crowd analysis. We discuss the main challenges in crowd counting and identify emerging trends, including integrating deep learning with attention mechanisms and combining Fuzzy logic with preprocessing techniques to achieve high counting performance in diverse and complex environments. Finally, we explore research strategies involving optimal hyperparameter selection and adaptive curriculum learning to enhance counting accuracy and overall system performance.

The remainder of this survey is organized as follows: Section 2 presents the literature review, Sect. 3 introduces the evaluation metrics, Sect. 4 describes the datasets, Sect. 5 reviews deep learning approaches, Sect. 6 discusses challenges in complex environments and highlights recent trends in crowd counting, while Sect. 7 concludes the paper.

2 Literature review

2.1 Comparative review of previous surveys

Recent research on crowd counting has evolved from classical image-processing techniques toward deep learning and lightweight deployment strategies. Early works laid the foundation by linking traditional density estimation with regression- and density-based models, while more recent surveys critically examined deep networks, attention mechanisms, offering taxonomies and clarifying trade-offs between annotation cost, scalability, and performance. Specialized studies addressed real-time and lightweight solutions, where methods such as compression-based networks, TinyCount, curriculum reinforcement learning, and knowledge distillation with multi-modal inputs demonstrated efficiency but still faced challenges in dense or cluttered scenes, training complexity, and data availability. Broader reviews further emphasized deployment on edge devices, highlighting opportunities but also gaps in benchmarking under real-world conditions. Building on these insights, our survey advances the field by integrating curriculum learning and fuzzy logic into preprocessing and deep learning pipelines, alongside hybrid attention mechanisms such as CANNet, EfficientNet, and VGG16. This combined strategy achieves stronger performance in complex scenarios, and as summarized in Table 2, offers clearer contributions compared to previous surveys in terms of focus, strengths, limitations, and practical relevance.

2.2 Crowd counting strategies

Crowd counting (CC) is about figuring out how many people are in a place using computer technology and artificial intelligence. It uses different methods; one identifies each person, another estimates the number from the image's features, and another creates a map showing where people are. Advanced techniques [32], involve using special types of neural networks to get better results. These methods are used in public safety, managing events, city planning, and understanding human behavior in stores. Researchers are constantly working to make these techniques more accurate and efficient; the methodologies can be illustrated in Fig. 2.

2.3 Traditional methods categories

2.3.1 Detection-based vs. heatmap-based approaches

Crowd detection methods, as discussed in [33], include both whole-body and partial-body approaches. Whole-body detection struggles with occlusions in dense crowds, which affects accuracy. Some studies, summarized below, explain various detection-based approaches used in crowd counting. It includes three methods: Haar wavelets, HOGs (Histogram of Oriented Gradients), and Support Vector Machines (SVM) explained in Eq. (1). Haar wavelets work well for whole-body detection in sparse crowds but struggle in dense crowds due to occlusion. HOGs allow for partial body-based detection by focusing on body parts like the head and arms, helping to handle occlusions. Support Vector Machines are mentioned as a detection-based method that also faces challenges with occlusion, but regression-based approaches are suggested as a potential solution to this limitation.

In the field of crowd counting, two main methodological approaches are commonly employed: detection-based approaches and heatmap-based approaches. Detection-based methods, such as YOLO or Faster R-CNN, focus on identifying and localizing people by drawing bounding boxes around people. These techniques are particularly effective in low-density crowd regions where individuals are more separated, and the issue of occlusion is relatively minor. One of the key advantages of detection-based approaches is that they not only provide the overall crowd count but also yield precise information about the location of.

In crowd counting, there are two main approaches: detection-based and heatmap-based methods. Detection-based methods, like YOLO and Faster R-CNN, detect each person with boxes. They work well in small crowds and give the exact location of everyone, but they struggle in very dense crowds. Heatmap-based methods create a density map instead of detecting each person. They work better in large, crowded scenes and can estimate the number of people even when individuals are hard to see, but they do not

Table 2 List of previous and state-of-the-art survey papers: pros, cons, and contributions to the scientific community

Ref/Year	Topic	Pros	Cons	Gap analysis
[17] 2019	General advances in density estimation via image processing	Broad overview; links traditional to deep learning methods; mentions applications	Limited dataset depth; lacks critical comparisons	Provides starting points for new researchers
[18] 2023	Algorithms for counting and density estimation in dense crowds	Explains transition from classical to DL; robustness in dense scenarios	Earlier surveys were only taxonomic, with little evaluation	Complement others by algorithm-level pros/cons
[19] 2023	Regression-based approaches	Explain the shift from detection to regression; historical context	Earlier surveys gave little detail	Highlights regression as a foundation of modern methods
[20] 2023	Deep networks applied to counting & density maps	Combines both perspectives; notes strengths/weaknesses	Earlier works separated them; they lacked a unified view	Provides integrated analysis with updated DL methods
[21] 2024	Learning paradigms for crowd counting	Shows trade-offs between annotation effort vs. performance; structured grouping	Previous surveys were biased toward supervised only	Adds by analyzing weakly/unlabeled data strategies and future directions
[22] 2024	Role of DL in crowd counting, specifically	CNN, GAN attention; emphasizes DL progress	Past surveys were general, not DL-specific	Clarifies DL's central role, challenges (domain adaptation, scalability)
[23] 2024	Lightweight dense crowd density estimation network with model compression	Strong memory efficiency with effective model compression	Compression may reduce the ability to capture subtle crowd details	Requires further testing on unconstrained real-world datasets to ensure reliability
[24] 2024	TinyCount: an efficient real-time CC network for intelligent surveillance	Extremely lightweight and optimized for real-time edge surveillance	May sacrifice accuracy in extremely dense or highly cluttered crowds	Need better trade-off strategies between efficiency and fine-grained accuracy
[1] 2025	DL methods for density maps + counting	Covers CNN, RNN, attention; and structured categorization	Earlier works lacked DL depth and comparison	Provides up-to-date taxonomy with pros/cons
[25] 2025	Comprehensive survey of crowd density estimation & counting	Broad coverage of classical + DL methods; datasets, losses, metrics included	Some redundancy with earlier surveys	It explains all the methods in one organized view
[1] 2025	Survey of DL methods for density estimation	In-depth algorithmic analysis of > 300 papers; structured taxonomy	Heavy focus on CNN methods; limited edge-case scenarios	It compares them clearly and shows pros and cons
[26] 2025	Crowd counting at the edge (knowledge distillation)	Highlights lightweight approaches; links edge computing with counting	Not a full general survey; more topic-focused	It shows if they can be used easily and don't waste much energy
[27] 2025	DL in image processing (general survey)	Wide coverage beyond counting (segmentation, detection, etc.); updated taxonomy	Less specific to crowd counting	It shows how deep learning works for other areas
[28] 2025	A comprehensive survey of lightweight neural networks for crowd counting	Provides a structured taxonomy and highlights trends in lightweight CC models	Lacks experimental benchmarking; no novel algorithm introduced	Missing a unified performance comparison across various datasets
[29] 2025	Lightweight dynamic convolutional network enhanced with curriculum reinforcement learning	Adapts dynamically to varying crowd scenes; curriculum RL improves convergence and robustness	Training process is complex, and reinforcement learning requires careful tuning	Needs large-scale validation on dense and highly diverse datasets to prove scalability

Table 2 (continued)

Ref/Year	Topic	Pros	Cons	Gap analysis
[30] 2025	Mutual head knowledge distillation framework for lightweight RGB-T	Efficiently fuses RGB and thermal modalities; distillation reduces computational burden	Depending on dual-modal input, which is not always available in real-world setups	Lack of lightweight single-modality models for adverse conditions
[31] 2025	Review of AI edge devices and lightweight CNN/LLM deployment	Provides insights into hardware–software co-design and real-world feasibility	Broad focus on AI deployment, limited attention to crowd counting specifically	It misses testing the models on small/real devices

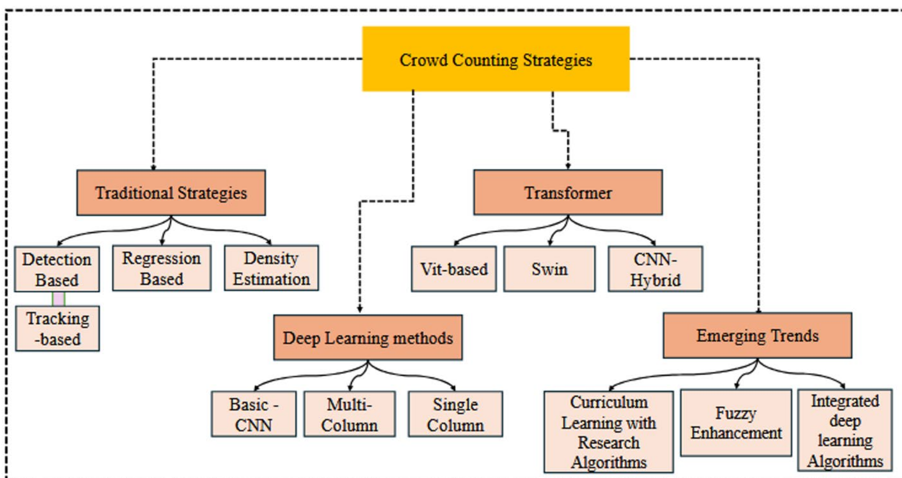


Fig. 2 Taxonomy of crowd counting strategies

show exact locations, which were introduced in [34], estimate a continuous density map or heatmap over the crowd image rather than attempting to detect everyone separately.

$$f\left(x\right) = w \cdot x + b \tag{1}$$

x is the feature vector extracted from the image, *w* is the weight vector learned during the training phase, *b* is the bias term, *f*(*x*) is the decision function of an SVM.

2.3.2 Regression-based approaches

Learn the regression of crowd image features to the crowd count using three steps: first, *foreground extraction*, which isolates important parts of the image, like the crowd, from the background. Second, there *is feature learning*, where meaningful features are extracted from the foreground mask to estimate the crowd count. Lastly, counting *regression features* involves studying regression-based approaches in CC, such as dynamic texture, wavelet analysis, and Gaussian regression. The dynamic texture approach improves occlusion handling but sacrifices accuracy in localizing individuals and understanding crowd distributions, while wavelet analysis and Gaussian regression are also widely used methods which proposed in [19, 35].

2.3.3 Density estimation-based methods
Estimate how crowded an area is instead of counting each person. Use heat maps to show the CD. It is good for very large and dense crowds; some studies have presented various density estimation-based methods for CC, emphasizing their effectiveness in large and dense crowds. The methods listed include MCCNN to capture features at different scales and combining conventional and dilated CNNs for enhanced receptive fields. TEDNet, which was introduced in [36] utilizes a trellis encoder-decoder network to improve spatial relationship accuracy in density maps, while HA-CCN is explained in [37], employs a hierarchical attention-based network to focus on relevant image regions.

2.3.4 Tracking-based methods
To enhance crowd counting systems, integrating Object Tracking is essential for monitoring individual movements across video frames. This approach, known as Tracking-by-Detection, involves detecting individuals in each frame and associating these detections over time to maintain consistent identities. Key tracking algorithms include: Kalman Filter: Predicts an object's future position based on its current state, effectively handling linear motion and occlusions. This filter can be introduced in [38]. Deep SORT enhances tracking by incorporating appearance features through a deep learning model, improving re-identification in crowded scenarios. This is discussed in [39].

Byte Track: employs a novel association method that considers all detection boxes, including low-confidence ones, to reduce identity switches and improve tracking accuracy. Integrating these tracking methods with detection-based approaches and enhancement processes, such as fuzzy, that were proposed in [40], enables not only accurate crowd counting but also detailed analysis of individual trajectories and behaviors, which is crucial for applications like surveillance and crowd management.

2.4 Deep learning strategies
These initial models created a foundation for future research but faced challenges with varying head sizes, uneven density distributions, and large perspective changes. As the field developed, newer CNN architectures were designed to tackle these issues more effectively, incorporating techniques such as multi-scale architectures, feature fusion, and attention mechanisms. Notable examples include MCNN [41] and CSRNet [42], which successfully applied these methods and achieved better counting accuracy in complex scenarios.

The second part focuses on more recent innovations in CC that go beyond traditional CNN approaches. Researchers are now combining CNNs with RNNs to improve temporal predictions and using transformers to gain a better understanding of the overall scene. GANs, as shown in [43], are also used to generate synthetic data to enhance accuracy. New strategies, such as weakly supervised learning and methods that require fewer labeled data, have made the models more efficient. Future trends may include 3D modeling, faster real-time processing, and better adaptability to different environments. Crowd-counting methods use deep learning models and can be grouped into several categories:

2.4.1 Basic CNN

A CNN is a DL model that excels at processing images by automatically learning to identify important features like edges and shapes. It works by using convolutional layers to scan the image and detect key details, pooling layers to simplify the data by focusing on the most important information, and fully connected layers to make a final prediction, such as counting the number of people in a crowd.

A conventional neural network is a deep learning model that processes images by automatically learning to detect important features like edges and shapes. In CC, CNNs work by first analyzing an image of a crowd through layers that highlight different details. Instead of directly counting people, many CNNs for CC produce a *density map calculated by* Eqs. (2) and (3). Each pixel in this map represents the estimated number of people at that location in the original image, as shown in [44, 45]. Finally, it sums up the values in the density map to give the total count of people in the image.

Figure 3 shows the steps for counting and estimating crowds. Traditional methods often have trouble counting people in very crowded scenes. To solve this, Wang et al. [9] introduced a deep regression network that works well in dense crowds by using a simple CNN to extract features efficiently. To make the method more robust, they also trained the CNN with images that had no people, giving them a count of zero. This reduced false alarms, and on the UCF-CC dataset, the method improved performance by almost 50% compared to a CNN trained without these negative samples [46].

Fu et al. [10] made an enhancement to crowd density estimation (CDE) by enhancing the network design and classification process. They removed redundant connections in the network using a similarity matrix, which made the network faster without reducing accuracy. They also added a two-stage classification: the first stage handles easy samples, and the second stage focuses on harder samples for more precise results. This method was tested on three datasets: PETS2009 (surveillance footage), Subway (subway scenes), and Chunxi Road (busy street), and showed significant improvements in both speed and accuracy of counting crowds.

$$D\left(x, y\right) = f_{CNN}\left(I\left(x, y\right)\right). \tag{2}$$

D (*x, y*) is the density map at pixel *(x, y) I (x, y)* is the input image at pixel. (*x, y*), f_{CNN} is represents the CNN model used to generate the density map.

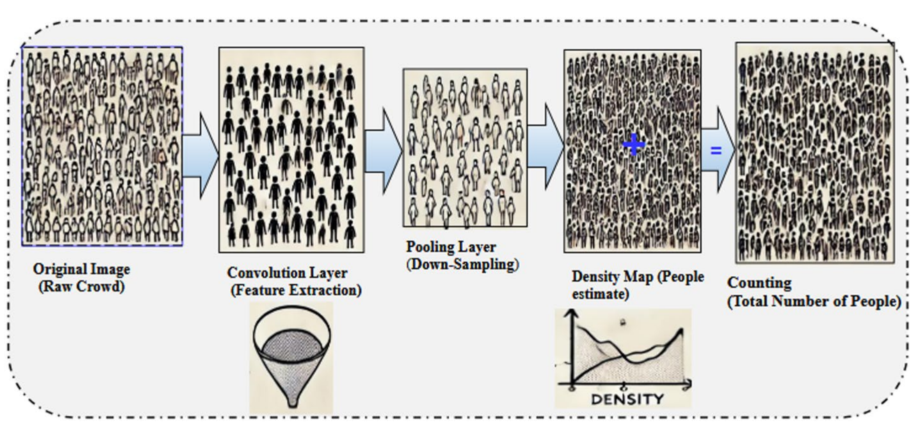


Fig. 3 Architecture of convolutional neural network for crowd density estimation

$$\hat{N} = \sum_{x=1}^{W} \sum_{y=1}^{H} D\,(x,) \tag{3}$$

$\widehat{N}$ is the estimated total count of people, (W, H) is the width and height of the density map $D\,(x, y)$.

$$L = \frac{1}{N} \sum_{i=1}^{N} \left| \widehat{N_i} - N_i \right| + \lambda \sum_{j=1}^{N} \left| \widehat{N_j} - 0 \right| \tag{4}$$

L is the loss function, $\widehat{N_i}$ is the predicted count for the $(i\text{th})$ image, N_i ground truth counts for the ith image, $\widehat{N_j}$ is the predicted count for the jth negative sample (should be close to zero), λ is a regularization parameter to control the importance of negative samples.

2.4.2 Multi column CNN (MC-CNN)

The MC-CNN uses multiple parallel columns to process images, capturing features like textures and edges. These features are combined in a fusion layer to produce detailed density maps, as shown in Fig. 4. Each column uses different filters to learn complex patterns, making predictions more accurate [41, 47].

The Relational Attention Network (RANet) [48, 49] applies self-attention (Eq. 5) to capture pixel relationships locally and globally, reducing noise and improving accuracy. In real-world crowds, densities vary, so DecideNet [34] combines detection and regression methods with an attention module to handle both low- and high-density areas effectively.

Crowd localization is enhanced by the Dilated Convolutional Swin Transformer (DCST) [31, 50], which integrates dilated convolutions Eq. (6) and transformers to capture broader context, improving counting accuracy even with occlusion and blur.

Although MC-CNNs are effective, they are complex and resource-heavy. Single-column CNNs (SC-CNNs) simplify training while maintaining good performance. The Double Multi-Scale Feature Fusion Network (DMFFNet) [51] uses VGG19 features, attention mechanisms, and a new loss function to create better density maps, outperforming previous methods.

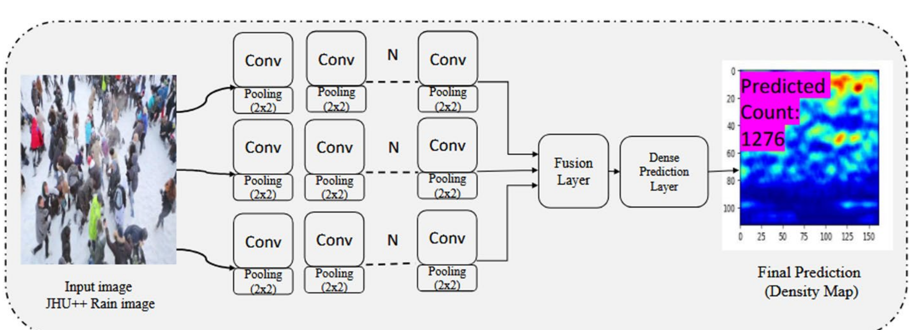


Fig. 4 Detailed structure of the multi-column CNN (MC-CNN) for crowd counting

$$A\left(x,y\right)=Softmax\left(\frac{Q\left(x,y\right)\cdot K{(x,y)}^{T}}{\sqrt{d_{k}}}\right)\cdot V\left(x,y\right) \tag{5}$$

Q (*x, y*), *K* (*x, y*), and *V* (*x, y*) is the query, key, and value feature maps, respectively. d_k is the dimension of the key vector, *A* (*x, y*) is the attention map.

$$y\left(m,n\right)=\sum_{i=1}^{M}\sum_{j=1}^{N}x\left(m+r\times i,n+r\cdot j\right)w\left(i,j\right) \tag{6}$$

The input *x* (*m, n*) has dimensions *M* and *N*, and the output *y* (*m, n*) is calculated.

using the convolution kernel *w* (*i, j*). The parameter r controls the dilation rate.

2.4.3 Single column CN

A SC-CNN is a neural network that uses just one column of layers to analyze images. This approach simplifies the network while effectively estimating CD and handling key issues in CC. SC-CNNs work by taking an image and passing it through one series of layers. First, convolutional layers scan the image to find features like edges. Then, pooling layers reduce the size of the image to make processing faster. After that, fully connected layers use the features to make predictions or classifications. This simple, single-path approach makes the network easy to use and quick to run. These steps can be drawn as in Fig. 5.

There are several networks used to improve single-column CNNs, such as the Congested Scene Recognition Network (CSRNet) can be highlighted in [52]. Hybrid approaches typically include methods that combine multiple techniques to enhance CC accuracy. For example, some approaches integrate multi-scale CNNs with density map regression to address varying person sizes and improve estimation accuracy. Others may merge density map methods with object detection models to benefit from both individual detection and overall density estimation. These hybrids are proposed in [53], aim to leverage the strengths of different techniques to handle scale variations more effectively and achieve high performance calculated by Eqs. (7) and (8) for mathematically calculating the density map and regression losses, as demonstrated in [45].

The Lightweight CC Network (*LCD net*) that was discussed in [54], presents a model for CDE in real-time video surveillance. It is designed to be lightweight and fast, so it can quickly analyze video frames without needing a lot of computing power. The model

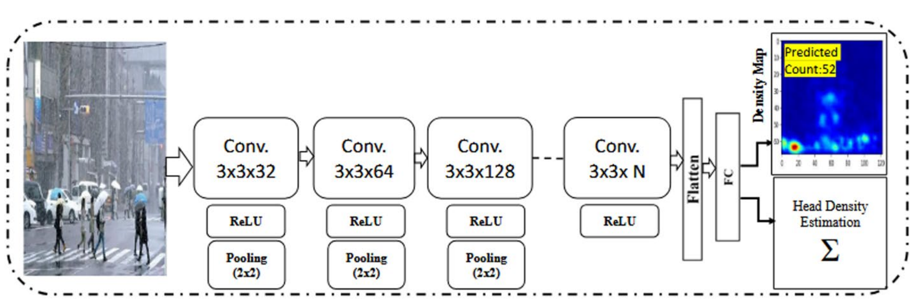


Fig. 5 Structural design of the single-column CNN (SC-CNN) for crowd counting

uses efficient methods to extract important details from images, allowing it to accurately measure CD even when crowds are very dense.

The authors in [55, 56], are introduced a Scale-Aware Adaptive Networks (*SANet*) presents a method that enhances crowd counting accuracy by integrating depth information into a CNN. The algorithm helps the network better handle varying sizes and distances of people in the crowd, improving its ability to estimate the count accurately. The methodology showed improved performance on standard CC datasets by effectively using depth data to manage scale variations and provide more counts that are precise.

$$F\left(x\right)=\sum_{i=1}^{N}\delta\left(x-x_i\right)*G_{\sigma i}\left(x\right),\sigma_i=\beta\bar{d}_i \tag{7}$$

N is the total number of individuals (points) in the image, x_i is the position of the ith individual in the image, $G_{\sigma i}\left(x\right)$ is the A Gaussian kernel with a standard deviation σ_i, δ is the ground truth function, β is the constant scaling factor, when equal to 0.3 gives the best performance, $\bar{d}_i$ is the Characteristic distance (such as from the camera) related to the ith individual.

$$\hat{D}\left(x\right)=g_i+\lambda d_i,g_i=\frac{1}{m}\sum_{i=1}^{m}log\left(1-D(G\left(Z^i\right))\right),d_i=\frac{1}{m}\sum_{i=1}^{m}log(D\left(X^i\right))+log\left(1-D\left(G\left(Z^i\right)\right)\right) \tag{8}$$

λ is the A weight balancing the GAN loss, m: number of samples, g_i is the UNet generator loss, $D\left(G\left(Z^i\right)\right)$ is the discriminator output, X^i is the represents real crowd samples from the training dataset, **N** is the number of training samples, $F_d\left(x_i;\theta\right)$. The predicted density map for the ith sample, $x_i$is the input ,θ is the represents the network parameters. D_i is the ground truth density map for the ith sample, $\|.\|^2$ is the squared Euclidean norm measures the difference between the predicted and actual density maps.

The Cascaded Pyramid Network *(CP-Net)* that shown in [57], presents a smart system for counting crowds in images. It works by first analyzing the image at different levels of detail, both close-up and far away.

The system refines crowd counts in several steps, combining detailed information to estimate people accurately in crowded and complex scenes.

The Hierarchical Attention-based Crowd Counting Network (HSNet) [58] handles people at different scales and focuses on key regions, improving accuracy even in dense or varied crowds.

Crowd counting is applied in areas like industrial monitoring, urban planning, and traffic management [59]. Traditional methods often ignore the effect of foreground and background. The Multi-Dense Scale Network (M-DSNet) addresses this by combining crowd counting with foreground-background segmentation and using dilated convolution and self-attention, achieving better results across four datasets.

The Scale Aggregation and Spatial-Aware Network (SASNet) [60], counts crowds from multiple camera views, keeping people at consistent scales, reducing background distractions, and combining all views for a more accurate count.

2.5 Transformer

Recently, attention mechanism-based models have shown remarkable success across various CV tasks such as introduced by Wang in [61]. The paper shows that transformer models, especially Swin Transformer, outperform CNNs in classifying static street-view building images, achieving about 90% F1-score and 92% accuracy, proving their strength in complex visual scene analysis. Rather than processing the entire image uniformly, these models enable the network to concentrate specifically on the most important regions, enhancing performance. In this context, crowd counting is a tough task in CV, often used in video surveillance and public safety. As camera resolutions increase and crowd images become more complex, accurately predicting crowd density and counting is more important than ever. Recent CNN-based methods have been effective in densely populated scenes, which were introduced in [62]. Also, a new approach to CC is introduced using an Encoder-Decoder Multi-Scale Attention Network. This approach builds on the strong ***U-Net*** architecture, enhanced with an attention mechanism. The multi-scale attention method is applied to different layers in the U-net, allowing the network to focus more on the crowd and less on the background. The attention mechanism, along with skip-connections, helps adjust the importance of various features while keeping information at different scales.

Recently, researchers have been improving crowd counting by using advanced deep learning methods. They are combining regular images with thermal data to handle tricky lighting and crowded scenes better. Attention mechanisms help models focus on important areas, and multi-scale processing lets them understand different crowd sizes. They're also creating lightweight models for faster, real-time predictions. To make models even stronger, they use techniques like GANs with U-Net, demonstrated in [63] for more accurate results to enhance the density map, which could be expressed as Eq. (9). *Curriculum learning* is used to teach models step by step, starting with easier tasks and moving to harder ones. Recent advances [64, 65] proposed that models are trained progressively with a loss function, which can be calculated from Eq. (10). These algorithms effectively handle various challenges, such as varying crowd densities and occlusions. They often utilize DL models with features like multi-scale processing and attention mechanisms to improve accuracy. Some methods focus on enhancing data efficiency, while others enhance spatial awareness. Overall, these approaches represent the latest advancements in accurately estimating crowd sizes in complex scenes.

$$\begin{aligned} \hat{D}\left(x\right) &= g_i + \lambda d_i, \quad g_i = \frac{1}{m}\sum_{i=1}^{m} \log\left(1 - D\left(G\left(Z^i\right)\right)\right). \\ d_i &= \frac{1}{m}\sum_{i=1}^{m} \log\left(D\left(X^i\right)\right) + \log\left(1 - D\left(G\left(Z^i\right)\right)\right). \end{aligned} \tag{9}$$

λ is the A weight balancing the GAN loss, m is the number of samples, g_i is the UNet generator loss, $D\left(G\left(Z^i\right)\right)$ is the discriminator output, X^i is the represents real crowd samples from the training dataset.

$$RAD\,(X,Y,D)_r = \sum_{i=x-r}^{x+r}\sum_{j=y-r}^{y+r} D_{i,j}$$
$$L\,(\theta)_{\text{step}} = \frac{1}{2N}\sum_{i=1}^{N}\left\|\text{AT}\,(D_i, step)\odot(X_i - D_i)\right\|_2^2 \tag{10}$$

r is a super parameter representing the local region size, D is the density map, D is the $G_\sigma(x)$, σ represents the spread parameter, $\mathrm{RAD}\,(\mathrm{X},\mathrm{Y},\mathrm{D})$ is the average density value of a rectangular region which takes (x, y) as the center, with a width of $2r+1$ pixels, θ is the parameters of a CC neural network, N is the number of samples in the training dataset, X_i, D_i ith is the input image and the corresponding ground truth, respectively.

2.6 Emerging trends

In recent years, several emerging directions have reshaped the taxonomy of crowd counting methods by introducing attention mechanisms, transformer-based designs, hybrid architectures, curriculum/adaptive training, lightweight models, and advanced preprocessing/fuzzy enhancements. For example, attention-driven models such as the Hierarchical Region-Aware Network (HRANet) employ region-aware and recalibration modules to enhance feature extraction in complex backgrounds and varying scales, significantly improving density map quality Xie et al. [66]. In the domain of transformer-based methods, Trans Crowd demonstrates how self-attention can effectively capture both local and global crowd patterns under weak supervision, using only count-level annotations instead of detailed head label Zhang et al. [14]. For lightweight and real-time applications, Tiny Count by Lee et al. [24] introduces an extremely compact network that achieves competitive performance on benchmark datasets, while being efficient enough for deployment on edge devices such as Raspberry Pi and Jetson Nano. Similarly, curriculum learning and adaptive training strategies, such as those presented in Highly Crowd Detection and Counting Based on Curriculum Learning, Fotia et al. [67], which gradually train models from sparse to highly dense crowd scenes, stabilize learning, and improve generalization to diverse environments. These trends illustrate how modern approaches are moving beyond traditional CNNs toward more adaptive, efficient, and robust solutions for crowd counting in complex real-world scenarios. In addition, fuzzy-based enhancements have recently gained attention for handling uncertainty and noise in complex environments. For example, Altundoğan et al. [68] introduced a Dynamic Fuzzy Cognitive Map (DFCM) approach for crowd analysis using time-series data extracted from video, showing improved robustness in noisy and low-quality scenarios. In addition to fuzzy-based preprocessing. we also added hybrid deep learning mechanisms, Elsepae et al. [69], employed to improve crowd counting in low-light and Harsh conditions, demonstrating the strength of combining different models for greater robustness.

3 Assessment metrics

In crowd counting, several evaluation metrics are used to measure how well a model estimates the population density. Here are the most common metrics, explained simply in both crowd counting and crowd-counting estimation in this paper [2].

1. Crowd Counting Evaluation [58]:

- MAE calculates the average difference between the predicted values and the actual values. In crowd counting, MAE is used to see how accurate the count of people is. If MAE is low, the count of the model is close to the actual number of people. The mathematical calculation of MAE can be expressed in Eq. (11).
- MSE is the square of the difference between the predicted and true counts, averaged over all test images. Like MAE, but it squares the differences before averaging them. MSE is used to evaluate how big the errors are in the crowd count or the density map. A lower MSE means fewer big mistakes. This error can be calculated from Eq. (12).
- RMSE is a way to measure how far a model's predictions are from the actual values. It works by calculating the difference between what the model predicted and the real answer, squaring those differences to make sure all errors are positive, and then averaging them. Finally, it takes the square root to bring everything back to the original scale. RMSE is useful because it gives more weight to large mistakes, meaning it highlights situations where the model was far off. A smaller RMSE means the model is doing a good job, while a larger one means it made bigger errors. RMSE can be calculated from Eq. (13).

2. Crowd Counting Estimation (CCE) [2]:

- Structural Similarity Index (*SSIM*) is a way to check how similar two images are. It looks at the important parts of the image, like shapes and patterns, instead of just checking individual pixels. In crowd counting, SSIM helps see if the predicted crowd map looks like the real one. A high SSIM means the two maps are very similar, and a low SSIM means they are different. SSIM can be evaluated from Eq. (14), whose value ranges between [− 1, 1].

Table 3 Definitions and representations of symbols used in Eqs. (11–15)

Symbol	Representation
$\boldsymbol{y_i}$	Actual headcount
$\widehat{\boldsymbol{y}}_{\boldsymbol{i}}$	Predicted headcount
ith	Test image
$\boldsymbol{\mu_x}, \boldsymbol{\mu_y}$	The average value for the first image and the second image, respectively
$\boldsymbol{\sigma_x^2}, \boldsymbol{\sigma_y^2}$	The standard deviation for the first image and the second image, respectively
$\boldsymbol{\sigma_{xy}}$	Represents how the values in the two images change together
$\mathbf{c}_1$	$(K_1L)^2$
$\boldsymbol{c}_2$	$(K_2L)^2$
$\boldsymbol{K}_1, \boldsymbol{K}_2$	$K_1 = 0.01, K_2 = 0.03$;are constant to prevent division by zero
x	The original image, y, is the generated image (like an image with added rain)
$\boldsymbol{\sigma_{xy}}$	$\mu_{i_r i_y} - \mu_{i_r}\mu_{i_y}$ (co-variation)
$\mathbf{MAX}_{\boldsymbol{i}}$	The maximum possible power of an image, x, y, is the original image and the generated image

- Peak Signal-to-Noise Ratio (PSNR) in CCE measures the quality of estimated density maps by comparing them to ground truth maps. It calculates the ratio of the maximum possible signal power to the noise power, affecting the accuracy of the estimated map. A higher PSNR value indicates that the estimated density map is closer to the true density map, reflecting a more accurate crowd count. PSNR can be estimated from Eq. (15). In addition, Table 3 explains the definition of every symbol.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{11}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y})^2 \tag{12}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i-1}^{n}(y_i - \hat{y}_i)^2} \tag{13}$$

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{\left(\mu_x^2 + \mu_y^2 + c_1\right)\left(\sigma_x^2 + \sigma_y^2 + c_2\right)} \tag{14}$$

$$PSNR(x,y) = 10\log_{10}\left(\frac{MAX_i^2}{MSE(x,y)}\right) \tag{15}$$

4 Datasets description

Several datasets have been created to help improve model generalization in the contemporary period. While early datasets contained low-density crowds, newer ones focused on high-density crowds, introducing challenges to CC. These larger datasets have driven the development of methods to address these issues. This section reviews eight key data sets, which play a crucial role in advancing algorithms by providing diverse scenarios for training and evaluation. Table 4 summarizes the types of the most famous datasets in the CC phenomenon. In addition to density-based datasets such as ShanghaiTech, UCF-QNRF, and JHU-CROWD++, crowd counting research can also benefit from detection- and tracking-based datasets. For example, CrowdHuman provides large-scale bounding-box annotations for pedestrians in crowded scenes, while CroHD (Crowd of Heads) and Cchead focus on head-level annotations and tracking information, making them particularly valuable for addressing challenges in highly congested environments. Using both density estimation and detection/tracking datasets allows crowd counting models to cover a wider range of crowd scenarios and improves their generalization.

When working with datasets like the *AI City Challenge* [70], annotation and generalization issues arise. In dense crowds, labeling is difficult due to occlusions, inconsistent

Table 4 Sample dataset descriptions for leading crowd counting benchmarks

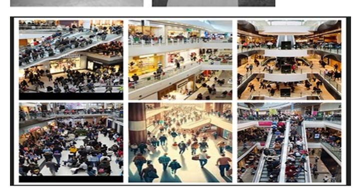
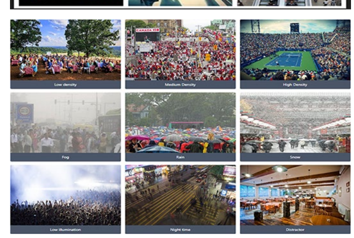
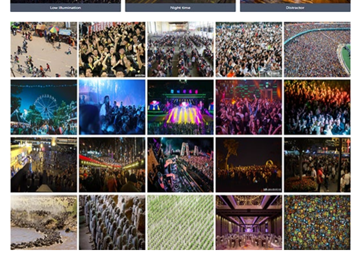
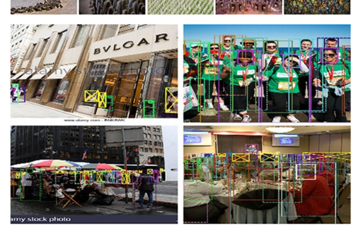
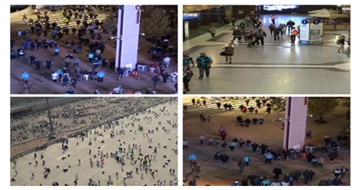
Datasets	Image quantity	Annotation quantity	Resolution	Best study	Samples of images
SHT (A), SHT(B)	1198	330,165	598 × 868	[79], 2020	
UCF-QNRF	1535	1,251,642	2013 × 2902	[80], 2023	
UCF-CC-50	50	63,974	2101 × 2888	[81], 2024	
UCSD	2880 video frames	49,885	158 × 238	[82], 2019	
Mall	2000 video frames	62,325	320 × 240	[45], 2023	
JHU-CROWD++	4360	1,438,808	Every image has a unique resolution	[4], 2020	
NWPU-Crowd	5109	2,133,238	Various resolutions, typically ranging from 640 × 480 to 1920 × 1080	[83], 2024	
CrowdHuman	15,000	330,000	1080 (varied)	[78], 2024	

Table 4 (continued)

Datasets	Image quantity	Annotation quantity	Resolution	Best study	Samples of images
CroHD (Crowd of Heads)	11,463 frames (9 video sequences)	2.27 million annotations, 5230 unique tracks	Full-HD	[84], 2024	
Cchead	50,528 frames (10 scenes)	2.36 million annotated heads, 2358 unique tracks	HD/Full-HD	[70], 2023	

or incomplete annotations, which can introduce noise. Bounding boxes may also fail to capture small or partially visible pedestrians.

Generalization across datasets is another challenge: models trained on one dataset often perform poorly on new environments or different crowd densities because of domain gaps. Fixed annotation strategies, such as Gaussian kernels, may not transfer well to other datasets. Additionally, lightweight models, although efficient, can struggle in complex crowd scenes. The crowd counting estimation, detection, and tracking datasets can be divided as follows:

a. Crowd Counting density map Datasets

1. ShanghaiTech (part A, part B): The ShanghaiTech dataset is a well-known benchmark for crowd counting, split into two parts to represent different crowd densities and scenes. Part A contains 482 images (300 for training and 182 for testing) with 241,677 annotated heads and mainly shows very dense outdoor crowds in urban areas such as streets and public squares. Part B includes 716 images (400 for training and 316 for testing) with 88,488 annotated heads and depicts sparse outdoor crowds in more relaxed places, such as parks. This dataset is widely used to develop and test crowd counting models, typically evaluated using MAE and MSE. The datasets referenced from the source in [47].
2. University of Central Florida Quantitative and Qualitative Crowd Counting Dataset (UCF-QNRF): The UCF-QNRF dataset is a large collection used for CC research, as mentioned in the source [71]. It contains 1535 images with 1,525,272 annotated heads, showing a wide range of crowd sizes from sparse to very dense. The images come from urban areas, indoor spaces, and events. The official split consists of 1201 training images and 334 test images, which has become the standard protocol in literature. It covers a wide range of crowd sizes, from sparse to extremely dense, and includes both indoor and outdoor urban scenes, such as streets, squares, and public events. This large-scale dataset is an important benchmark for testing the scalability of crowd counting methods.
3. University of Central Florida Crowd Counting 50 (UCF-CC-50): The UCF-CC-50 dataset is used for CC and has 50 images with 63,296 annotated heads. It focuses

on very crowded scenes in urban areas. Even though it is a smaller dataset, it is useful for testing and improving CC models, with performance usually measured by errors like MAE and MSE. It focuses on extremely dense outdoor crowds in urban areas such as streets and plazas. For UCF-CC-50, there is no official split. The standard protocol is 5-fold cross-validation, where 40 images are used for training and 10 for testing in each fold, and results are averaged over all five folds. These datasets can be found in [72].

4. University of California, San Diego (UCSD): The UCSD dataset is used for CC and consists of 2880 video frames with 7014 annotated people. It focuses on sparse to moderately dense crowds like sidewalks and streets in outdoor settings. The dataset features continuous video sequences, which help analyze crowd movement and dynamics. It features sparse to moderately dense outdoor crowds, typically along sidewalks and campus walkways. Because it provides continuous video sequences, it is also used to analyze crowd movement and temporal dynamics in addition to still-image counting. There is no official split; a common protocol uses 800 frames for training and 1200 for testing. The source of the datasets is [73].
5. Mall: This contains 2000 video frames with about 60,000 annotated pedestrians, captured from a fixed surveillance camera in an indoor shopping mall. It depicts moderate to high crowd densities in areas such as walkways and food courts, such as [74]. A common protocol uses the first 800 frames for training and the remaining 1200 for testing. It represents indoor mall environments, such as shopping areas and food courts, with moderate to high crowd density. This dataset provides valuable data for analyzing crowd behavior in enclosed spaces and for testing crowd counting models in indoor scenarios.
6. Johns Hopkins University (JHU-CROWD++):

   The JHU-Crowd + + dataset is an extended version of the original JHU-Crowd dataset, created for crowd counting (CC). It contains 4372 images with annotations for over 1.51 million people, covering a wide range of crowd densities, from sparse to extremely dense. The images include scenes such as streets, public events, and gatherings, providing diverse examples to train and evaluate CC models. This extended version offers even more challenging crowd scenarios for training and testing. The dataset is divided into three subsets: 2272 images for training, 500 for validation, and 1600 for testing. This standard split ensures consistent benchmarking and allows fair comparisons among different crowd-counting methods. The dataset can be accessed from the official source in [4].
7. Northwestern Polytechnical University (NWPU-CROWD):

   The NWPU-Crowd dataset is designed for crowd counting (CC) and contains 5109 images with annotations for over 2 million people. It covers a wide range of crowd densities, from sparse to extremely dense, and includes various environments such as streets, parks, and large public events. This diversity makes it one of the most comprehensive datasets for evaluating crowd counting algorithms. The dataset is split into 3109 images for training, 500 for validation, and 1500 for testing, providing a standardized setup for fair and consistent benchmarking. The dataset can be accessed from the source in [75].

b. Crowd Counting density map Datasets

1. The CrowdHuman: The CrowdHuman dataset is designed for detecting people in crowded V scenes. It contains about 15,000 images with over 330,000 labeled people; each annotated with a bounding box and an occlusion level indicating how much others block a person.
   Most images are outdoor scenes from streets and public places, showing people from different angles, lighting conditions, and crowd densities. On average, each image has around 22 people, with some images containing more than 100. The dataset is divided into 10,000 images for training, 4370 for validation, and 5000 for testing, enabling models to be trained, tuned, and evaluated effectively. While the dataset mainly focuses on outdoor scenes, some images include semi-indoor public areas. Its main goal is to capture realistic crowded environments with varied lighting, angles, and occlusions. The CrowdHuman Dataset can be accessed from the official website [76].
2. Crowd of Heads dataset: The CroHD dataset is designed for tracking pedestrian heads in dense crowds. It contains 9 video sequences with 11,463 frames captured from elevated viewpoints at 25 fps. The dataset includes both indoor and outdoor scenes, such as train stations and pedestrian crossings, under varying lighting and conditions. Each frame is annotated with head bounding boxes, totaling over 2.27 million annotations and 5230 unique tracks. The official split uses 4 sequences for training and 5 for testing. CroHD focuses on head-level tracking to handle occlusions and overlapping people, and introduces the IDEucl metric for evaluating identity preservation. It is publicly available for research. The datasets stored in [77].
3. Chinese Large-scale Cross-scene Pedestrian Head Tracking; The Cchead dataset [78] is a large benchmark for tracking pedestrian heads in dense crowds. It has 10 scenes with 50,528 frames, over 2.36 million annotated heads, and 2358 unique tracks. The scenes include pedestrians moving at different speeds and directions, captured from slope and overhead viewpoints.
   The dataset is split into 40,422 frames for training and 10,106 for testing, supporting proper model training and evaluation. It mainly covers outdoor areas, like streets and school roads, and is publicly available for research.

5 Deep learning techniques for solving crowd counting problems

a. Related works

In this section, Table 5 compares the performance of crowd counting algorithms across different datasets (ShanghaiTech Part A/B, UCF-QNRF, and UCF-CC-50). The reported MAE, MSE, and RMSE values highlight how each method adapts to problems of crowd counting, such as scale variation, perspective distortion, uneven distribution, rotation, occlusion, and bad weather.

To solve the problem of scale variation, Li et al. presented an algorithm, AP-FPN, which demonstrated strong robustness with relatively low error (MAE = 54.9 on SHT A). Zhang et al. introduce a Segmentation-based CC Network that improved robustness in bad weather through segmentation, albeit at a higher computational cost. To get rid of

Table 5 Benchmarking crowd counting models: performance metrics across diverse challenges

Ref Year	Algorithm	Challenge	Metrics MAE/MSE SHT A	SHT B	UCF-QNRF	UCF-CC-50	Pros	Cons
[85] 2020	MSA-CNN	Scale Variation	72.4 104.7	22.7 35.4	293.9 361.6	–	Robust to scale variation; good accuracy on dense scenes	Limited with uneven distribution and background noise
[86] 2020	ALN	Perspective Distortion	63.5 99.2	8.1 11.9	110.3 178.2	211.4 306.7	Adapts well to perspective distortion	Weak performance in very high-density scenes
[87] 2020	DM-CC	Uneven Distribution	59.7 RMSE/95.7	7.4 11.8	85.6 148.3	211 291.5	Handles uneven crowd distribution effectively	Sensitive to noise and occlusion
[63] 2021	Segmentation-based CC Network	Bad weather Scale Variation	68.3 104.1	12.1 19.3	–	233.6 352.6	Incorporates segmentation to improve under bad weather	Higher computation; less accurate in open scenes
[88] 2021	AP-FPN	Scale Variation	54.9 85.3	7.4 10.8	75.7 120.4	186.8 266.1	Strong handling of scale variation with pyramid features	High memory and computational cost
[59] 2021	DSNet	Scale Variation	61.3 RMSE/97.3	6.7 10.5	91.4 160.4	186.8 266.1	Reduces counting error across multiple scales	Weak under complex backgrounds
[89] 2021	SACCN	Complex Background	59.2 98	6.8 10.5	178.0 258.5	96.1 167.8	Attention improves performance in cluttered scenes	Training complexity; sensitive to hyperparameters
[90] 2022	CSS-CCNN	Rotation	–	–	437.0 722.3	564.9 959.4	Robust for dense crowds using a cascade structure	Slower inference
[91] 2022	FPN, Auto-scale	Scale Variation	75.7 150.4	10.4 18.8	124.8 234.7	224.6 314.6	Flexible for scale variation	Accuracy dropsin very dense scenes
[92] 2022	SGANet	Scale Variation	58.0 100.4	6.3 10.6	89.1 150.6	–	GAN improves density map quality	Prone to instability in training

Table 5 (continued)

Ref Year	Algorithm	Challenge	Metrics MAE/MSE SHT A	SHT B	UCF-QNRF	UCF-CC-50	Pros	Cons
[93] 2022	SDC	Scale Variation	69.9 106.9	9.4 15.9	–	–	Effective for uneven distribution using local context	Limited for dynamic large-scale variations
[94] 2022	CNN + CBAM	Scale Variation	64.6 103.2	8.3 14.1	113.8 188.6	–	Attention enhances feature representation	Increased model complexity
[95] 2022	2U-Net	Rotation Scale Variation	63.3 103.8	7.4 12	239.4 356.1	1.5 1.9	Robust to rotation; reliable in multi-scale scenarios	Weaker under heavy occlusion
[96] 2022	HRANet	Uneven Distribution	100 87.2	7.2 9.7	84.6 146.2	160.9 235.8	Hierarchical attention improves performance	Requires large training datasets
[97] 2023	Deep Temporal CC	Uneven Distribution	60.8 100	7.2 8.3	88.7 93.2	319.9 312.6	Temporal modeling improves video counting	Not suitable for still images
[98] 2023	CAFNet	Scale Variation	97 94.5	10.2 12.2	96.4 155.5	–	Fuses multi-scale and distribution awareness	Computationally expensive
[99] 2023	FPANet	Scale Variation	70.9 120.6	6.9 15.5	108.9 197.6	159.5 218.4	Accurate with feature pyramid attention	Sensitive to cluttered backgrounds
[100] 2023	AWCC-Net	Bad Weather	56.2 91.3	–	76.4 130.5	–	Good robustness to weather-related noise	Limited generalization to unseen conditions
[101] 2024	EfficientNet-B3	Rotation	107.6 61.3	11.2 6.25	–	213.2 162.3	Lightweight with good accuracy	Rotation handling is less stable under high occlusion
[102] 2024	Detailed architecture	Scale Variation, Rotation	–	–	97.20 156.4	201.6 286.4	Accurate regression-based counting	Limited interpretability
[103] 2024	DCNN	Occlusion Perspective Distortion	52.6 90.9	8.1 12.8	96.4 168.7	181.8 240.6	Strong against occlusion and perspective distortion	Heavy model; higher training cost

Table 5 (continued)

Ref Year	Algorithm	Challenge	Metrics MAE/MSE SHT A	SHT B	UCF-QNRF	UCF-CC-50	Pros	Cons
[104] 2024	Hybrid Lightweight Network	Scale Variation	71.5 108.6	11.3 20.0	100.4 182.6	–	light-weight, efficient, lower annotation cost, and competitive accuracy	weaker in very dense crowds, limited generalization, and less robust under occlusion.
[24] 2024	Tiny Count	Real-time crowd counting	78.2 120.8	10.81 18.4	134.7 223.3	–	Very light-weight; fast inference on Raspberry Pi & Jetson devices	May lose accuracy in ultra-dense/ occluded datasets
[105] 2024	TRepVGG	Scale variation; occlusion in dense crowds	66.9 104.5	7.4 12.5	–	–	Light-weight and efficient; faster inference; suitable for real-time edge use; reduces training	Accuracy is slightly lower than larger transformer-based models; it needs careful tuning for reparameterization
[106] 2024	MPCount	Domain generalization	115.7 199.8	11.4 19.7	165.6 290.4	–	Improves robustness when training and testing on different datasets	May under-perform in highly diverse unseen domains
[107] 2024	Occupancy Counting at Entrances	Real-time Dense Entrance Monitoring	Head-tracking-for-occupancy-counting Accuracy:98.1%				High accuracy for smart building applications	Task-specific; limited generalization to large open scenes
[108] 2025	DMLViT	Handling heavy congestion, scale variation, and occlusion.	53.1 86.5	6.2 9.7	85.6 148.3	211.0 291.5	Captures local + global features; robust to scale variation; effective in highly congested traffic scenes; improves transformer efficiency	Higher computational cost than light-weight CNNs; may need powerful GPUs for training

Table 5 (continued)

Ref Year	Algorithm	Challenge	Metrics MAE/MSE SHT A	SHT B	UCF-QNRF	UCF-CC-50	Pros	Cons
[109] 2025	D2PT	Scale variation; occlusion in dense crowds	68.2	77.9	58.1	–	Improves both localization and counting; combines the benefits of density and point methods	Likely heavier/ training complexity
[110] 2025	P2PNet	Occlusion	52.74 85.06	6.25 9.90	85.32 154.50	-	Data-efficient, much faster than P2PNet	Sensitive to parameters, weaker with motion blur, and depends on pseudo-label quality
[111] 2025	CrowdCL	Domain adaptation, Scale variation	142.5 RMSE:231.4	150.7 243.9	–	–	No need for many labels Works better on new datasets	Less accurate than supervised models, still hard with heavy occlusion
[112] 2025	PPSC	Dense crowds with perspective distortion	62.6 96.6	7.9 13.1	86.3 138.9	–	Need fewer labeled images Use perspective to improve counting. Handles scale changes better	Less accurate than fully supervised methods Needs perspective maps Struggles with heavy occlusion
[113] 2025	SACCN		52.4 84.3	6.5 10.7	77.2 125.0	202.3 280.5	Effectively handles multi-scale crowd density	Increased model complexity and computation
[114] 2025	MISF-Net (RGB-T Fusion)	Modality fusion (RGB + Thermal)	DroneRGBT: Datasets MAE:10.80 RMSE:19.87				Robust under poor illumination and nighttime scenes	Requires dual-modal data (not always available)
[115] 2025	MSFFNet	Scale variation & semantic context	58.6 93.2	6.2 10.1	85.3 146.7	180.2 245.0	Captures both global and local features effectively	Higher computational cost due to multi-branch fusion

uneven distribution, Wang et al. proposed a DM-CC that achieved reliable performance but remained sensitive to noise and occlusion.

Attention-based methods became dominant with Chen et al. when they presented HRANet, which achieved the lowest error (MAE = 52.8, MSE = 87.2 on SHT A) and demonstrated strong adaptability to uneven crowd distributions and complex backgrounds. Similarly, Liu et al. presented CNN integrated CBAM enhanced feature representation through attention modules, though at the cost of increased model complexity.For temporal and multi-scale modeling, Zhou et al. introduced a method, Deep Temporal CC improved performance on video sequences. While Xu et al. put forward FPANet leveraged pyramid attention to enhance precision under scale variation. Tan and Le suggested an EfficientNet-B3, which provided a lightweight but accurate solution. Khan et al. gave a study, Crowd-Regression Module enhanced regression-based interpretability across datasets but faced limited robustness under occlusion.

More recent works emphasized lightweight and transformer-based models. Huang et al. introduced a Hybrid Lightweight Network reduced annotation costs while retaining competitive accuracy, though performance dropped in very dense crowds. Zhu et al. suggested a study of TinyCount achieved real-time crowd counting on edge devices such as Raspberry Pi, highlighting efficiency but limited scalability. Transformer-driven models like Gao et al. as DML-ViT model and Feng et al. D2PT algorithm improved handling of congestion, scale variation, and occlusion, albeit with heavier training complexity. Finally, Zhang et al. introduced domain adaptation with reduced label dependency, though accuracy lagged fully supervised approaches.

Despite these advancements, most existing crowd counting approaches still face several limitations. Many models show reduced accuracy in extremely dense and highly occluded scenarios, where individuals overlap significantly. Others struggle to generalize well across unseen datasets, indicating limited adaptability to different environments and lighting conditions. In addition, transformer-based and attention-driven methods often require heavy computational resources and long training times, making them less practical for real-time deployment on low-power devices. Lightweight models, while efficient, usually sacrifice accuracy when applied to complex or large-scale crowd scenes. These challenges highlight the need for more robust, scalable, and computationally efficient solutions.

b. Practical analysis

In this survey, the focus is on providing a comprehensive overview rather than detailing a specific methodology. Therefore, the explanation of the methodology is not as in-depth as it would be in an experimental study. Essentially, I'm presenting a broad survey that highlights existing approaches and their strengths and weaknesses.

From Table 4, the results were good in sparse scenes, but the error was higher in dense scenes. To solve this problem, the practical analysis achieves balanced results in both high-density and low-density crowd scenarios by combining several complementary techniques. First, fuzzy-based preprocessing with sharpening and Laplacian filtering enhances contrast and reduces noise, which makes head regions clearer in low-light or

Table 6 Our proposed experiments for crowd counting based on new trends

Algorithm	Datasets	Challenge	Metrics	Technique	Pros	Cons
EffiVGGNet [69] VGG16	SHT A SHT B/ SHT A SHT	Harsh Weather	MAE: 77.7028 MAE: 12.9224/ MAE: 51.8785 MAE: 1.9903	Hybrid deep learning mechanisms, EfficientNetB7, concatenated with VGG16	Combines the strengths of EfficientNet (efficiency) & VGG (stability); good for harsh weather	More parameters, so higher computational cost
CANNet_+ CL	SHT A SHT B JHU++	Perspective & Complex Background	MAE:42.5 MAE:17.24 MAE:69.63	Applied adaptive curriculum learning with attention mechanisms	Improves learning gradually; robust to complex background	Longer training; sensitive to curriculum design
MC-CNN	SHT A SHT B UCF-QNRF	Rainy Weather	MAE:59.5 MAE:16.7 MAE:102.99	Hybrid Attention mechanisms with MC-CNN architecture	Good robustness against noise/weather	Less effective under heavy occlusion
CSRNet	SHT A SHT B UCF-QNRF	Scale Variation	MAE:96.20 MAE:29.20 MAE:66.40	CSRNet with channel and spatial attention mechanisms	Strong for scale variation; high accuracy on benchmarks	not robust for background clutter
Bidirectional LSTM	HAJJv2 Dataset	Dense crowd, uneven distribution	mAP@50: 95.5 Precision: 94.8% Recall: 93.1%	Processing by sharpening and Laplacian, and fuzzy based on BiLSTM	strong in dense scenes	Needs sequential data
YOLO8n	video crosswalking/YOLO5-Deepsort crowd datasets	Illumination & Occlusion	mAP@50: 97.7% Precision:97% Recall:96.8%/ mAP@50: 98.7% precision:97.9% Recall:96.5%	The datasets were enhanced by CLAHE and gamma correction based on YOLO11n	Excellent detection speed and accuracy	Sensitive to extreme occlusion; requires tuning for light changes

uneven illumination conditions. Next, hybrid attention mechanisms, integrating both channel and spatial attention, guide the model to focus on the most relevant features and locations, improving robustness against scale variation, occlusion, and background clutter. The fusion of different CNN architectures, such as EfficientNet and VGG-style networks, further strengthens performance by capturing both fine details and global context, enabling accurate recognition across varying crowd densities. In addition, curriculum learning is applied to train the model gradually, starting from simple low-density cases and progressing to complex dense and occluded scenes, which stabilizes training and improves generalization. Also, YOLOv8n provides a practical balance between performance and speed, making it particularly suitable for scenarios that require deployment on edge devices or in resource-constrained environments. As mentioned in [116], which proposal is a system for human density monitoring that combines YOLOv8 for

Table 7 Challenges of crowd counting in complex environments

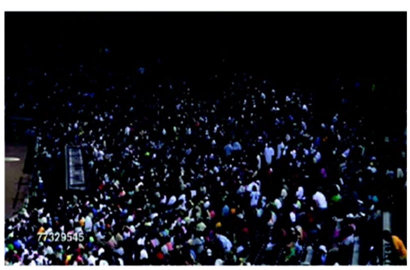
Problem	Scene	Proposed Solution	Techniques	Strengths/ weaknesses
1. Occlusion Where parts of the crowd are blocked from view, making it hard to detect individuals		• CSONet (Characterizing Scattered Occlusion Network) [117] • Multiscale feature extraction module [33]	• CNN with attention + scattered features • Multiscale CNN filters	• Strong in complex scenes Overestimates sometimes, complex • Very accurate, simple, and stable • May miss small, occluded details
2. Complex background Background objects may look like people, making counting harder		• SACANet (Scale-Adaptive Context-Aware Network) [118] • Scale-Aware + Regional/Semantic Attention [59]	• CNN + FPN (Feature Pyramid Networks) + Attention • SANet, RAM, SAM Modules	• Accurate on both sparse and dense scenes, robust to distractions • Can struggle on extremely crowded datasets, with higher error penalties
3. Rotation Viewpoint and rotation changes affect crowd visibility and estimation		• DCNN + RNN with LSTM (Global Feature Embedding + RSAR) [119] • Real-time CNN-based estimation with CLBP & LR-CNN [120]	• CNN + LSTM (for spatial-temporal learning) • CSRNet, CAN, CLBP features, MSE loss, Adam optimizer	• Performs well in varying angles and less dense scenes • Struggles with very dense scenes • Strong real-time application and safety enhancement
4. Variation in illumination Low-light conditions reduce image quality, making CC difficult		• Low-light enhancement with LE-Curve, Ct-Net, and CNN [121] • RGB-T fusion using dual-level alignment (Crowd Align) [122]	• Histogram Equalization, Adaptive Gamma Correction, DL-based RLEM, HCA, CNN • Shared-weight backbone, DSPA, LFAF, Regression Head	• Enhanced low-light performance (MSE: 26.53 on RGBT-CC) • Effective in thermal-RGB fusion • May depend heavily on preprocessing and thermal image quality

Table 7 (continued)

Problem	Scene	Proposed Solution	Techniques	Strengths/weaknesses
5. Uneven distribution Variation in crowd density and head size across the image makes accurate counting difficult		• Swin Transformer with Feature Adaptive Fusion Head (FAFHead) [123] • Frequency domain-based density map learning [124]	• Patch-based Transformer for inter-region context • Adaptive fusion for handling head size & CD • Loss function via characteristic function in the frequency domain	• High accuracy on varied-density datasets • Efficient training and better generalization • Complexity in Transformer computation and frequency domain methods
6. Scale variation Scenes with large variations in head sizes due to different distances from the camera		• Hierarchical Dense Dilated Deep Pyramid Feature Extraction (HDPF) [93] • Deep Pyramid Feature Extraction Module (PFEM) • A CNN model that uses attention mechanisms (CBAM) + VGG16 feature extraction [94]	• PFEM with densely connected dilated convolutions • VGG-16 for feature extraction • Convolutional Block Attention Module (CBAM) • Specialized conv + deconv layers for density map creation	• Effectively handles scale variation, reduces sparse sampling, and redundant features • Potential computational complexity
7. Perspective distortion When people in a crowd photo look bigger, if they are close to the camera and smaller if they are far away. This size difference makes it hard for models to count people accurately		• Adaptive learning-based (CAL) perspective correction before CNN-based density estimation [86] • PACNN (PA-Net) Integrates perspective maps with CNN using a perspective-aware loss (PA-Loss) [125]	• Adaptive distortion correction • Perspective-aware loss (PA-Loss) • Perspective map integration • Specialized dual-objective loss function	• Significantly improves MAE and MSE vs. non-corrected methods • Maintains reasonable inference speed (12 FPS) • Handles scale/density variation well

Table 7 (continued)

Problem	Scene	Proposed Solution	Techniques	Strengths/weaknesses
8. Changes in the Weather Rain, fog, and sunlight reduce image quality and affect detection accuracy		• AWCC-Net: Weather-aware CC [63] • Pix2Pix GAN-based denoising + CNN [126]	• Multi-scale CNNs, GANs, and Transformer decoders • Weather query and adaptive attention • Pix2Pix GAN for image cleanup before CC	• AWCC-Net achieves good accuracy across weather conditions • Pix2Pix GAN improves PSNR & SSIM significantly • Performance can vary with real-world noise; models may need retraining for new weather types

detection and DeepSORT for tracking. It achieves accurate real-time crowd analysis from aerial views but faces challenges in very dense or highly occluded scenes.

Hybrid strategies reduce errors in dense crowds while maintaining high accuracy in sparse scenes in images and videos, also leading to a more reliable and adaptable framework for real-world crowd counting challenges. Can be shown in the following Table 6. The tools and libraries employed in the experiments were clearly specified. MATLAB was used for simulation and analysis, while Kaggle and Google Colab were used for model training and testing. Statistical visualizatione was carried out using Matplotlib, Seaborn, Plotly, and Stats models, and fuzzy modeling was performed with scikit-fuzzy and ANFIS. These tools were integrated across preprocessing, training, evaluation, and visualization to strengthen the survey with practical insights and ensure transparency and reproducibility.

6 Challenges of CC in complex environments

Table 7 addresses eight key challenges in crowd counting, such as occlusion, cluttered backgrounds, varying viewpoints, low light, uneven densities, scale variation, perspective distortion, and bad weather. In addition to the solutions include DL techniques like attention mechanisms, context-aware networks such as SACANet, fusion of thermal and RGB images, transformer models, and weather-adaptive CNNs to improve counting accuracy.

Table 8 highlights recent trends in AI-based crowd counting, including the use of attention mechanisms, self-supervised learning, multi-feature fusion, and lightweight models for real-time use. Training with unlabeled data is also gaining attention, improving model accuracy and efficiency while reducing labeling costs. To improve both the crowd count as well as the density map generation, future work and recent studies have been introduced as several advancements to address existing challenges. Graph neural networks (GNNs) have been used to model complex spatial and temporal dependencies, while adaptive multi-scale learning and curriculum learning approaches improve model adaptability across varying crowd densities. Self-supervised and semi-supervised learning methods reduce the dependence on large, labeled datasets, making training more

Table 8 Advancements in CC: proposed objectives and new directions

Study	Objective	New trend
[127] 2020	Proposes using graph neural networks (GNNs) combined with other methods to better model complex relationships and interactions in crowded scenes, improving CC accuracy	The latest trend involves integrating GNNs with DL techniques and attention mechanisms to enhance the modeling of spatial and temporal dependencies in CC
[128] 2021	The paper introduces a method to accurately count people in highly crowded areas by using adaptive multi-scale context learning, which allows the model to adjust to different crowd densities and patterns for better results	Combined Transformer models to better capture details in dense crowds, with semi-supervised learning to reduce the need for large, labeled datasets, improving performance across various environments
[129] 2022	Using self-supervised learning. This allows the model to learn features from unlabeled data	Emerging trends include domain adaptation and contrastive learning to further boost cross-domain performance, enabling models to handle varying conditions with minimal labeled data
[94] 2022	The article focuses on using an attention-aware CNN network to improve CC in challenging environments. It enhances the model's ability to focus on important areas and details within complex scenes, leading to more accurate crowd estimates	integrating attention mechanisms with CNNs to better handle complex scenes and varying crowd densities, improving the model's ability to focus on critical regions and details in dense or occluded environments
[130] 2023	Aims to improve CC accuracy by combining multi-scale feature **fusion** with attention mechanisms. This approach enhances the model's ability to capture and utilize features at various scales and focus on important areas within crowded scenes	The new trend involves integrating multi-scale feature fusion and attention mechanisms to better handle varying crowd densities and complex scenes, leading to more precise CC by capturing detailed and relevant information effectively
[67] 2023	The paper uses curriculum learning to improve crowd detection and counting by progressively training on simpler to more complex crowd scenarios.	The trend is applying curriculum learning to enhance model performance in dense crowd situations by gradually increasing training difficulty
[131] 2024	The used architecture for countig is Single-column CNNs	The trend is toward using lightweight and efficient architectures like Mobile Net or Efficient Net in CC to achieve high performance with reduced computational resources and complexity
[132] 2024	The paper focuses on improving CC accuracy in challenging weather conditions by using multi-queue contrastive learning. This technique helps the model distinguish between different features and enhance its robustness against adverse weather effects	Leveraging contrastive learning and multi-queue learning techniques are used to improve model performance under difficult conditions, such as adverse weather, by better differentiating and learning from diverse feature representation

efficient. Attention-aware CNNs and multi-scale feature fusion with attention mechanisms enhance the ability to focus on important regions and handle dense or occluded crowds. Additionally, lightweight and efficient architectures, such as MobileNet, have been proposed to reduce computational cost without sacrificing performance. More recent works also highlight the importance of domain adaptation and contrastive learning for handling cross-domain variations, as well as multi-queue learning to improve robustness against challenging conditions like adverse weather.

7 Conclusion

Crowd counting has evolved from limited traditional approaches into a mature research area powered by deep learning. Convolutional neural networks and advanced architectures now enable accurate estimation under diverse densities and challenging real-world conditions. This survey not only reviews existing methods but also provides a structured comparison of their strengths and limitations, offering clear insights into how different approaches perform across varying scenarios. Furthermore, it highlights critical research gaps and emerging strategies, giving researchers a roadmap to develop more reliable and adaptable models.

The next stage of research must emphasize scalability, robustness, and real-time deployment. Exploring novel architecture, optimizing model parameters, and validating unseen datasets will be key to ensuring generalization. In addition, extending applications to live crowd monitoring and surveillance will bridge the gap between theory and practice. By focusing on adaptability and efficiency, future systems can deliver reliable crowd analysis, making them essential tools for safety, planning, and intelligent public space management.

Acknowledgements
Funding Open access funding provided by The Science, Technology & Innovation Funding Authority(STDF) in cooperation with The Egyptian Knowledge Bank (EKB). Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowl-edge Bank (EKB). The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Author contributions
All the authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by Heba F. Elsepae and Ehab K. I. Hamad. The first draft of the manuscript was written by Heba F. Elsepae, Heba M. El-Hoseny, and El-Sayed M. El-Rabaie commented on the previous versions of the manuscript. All the authors read and approved the final manuscript.

Data availability
Data is provided within the manuscript.

Declarations

Ethics approval and consent to participate
This study did not involve any human participants, animal subjects, or clinical data. Therefore, ethical approval was not required according to institutional and international guidelines. No human subjects were involved in this research. Consequently, consent to participate was not applicable.

Consent for publication
This manuscript does not contain any person's data in any form (such as images, videos, or identifying personal information). Therefore, consent for publication was not applicable.

Competing interests
The authors declare no competing interests.

Funding
Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB). Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB). The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Received: 29 June 2025 / Accepted: 8 January 2026

Published online: 17 February 2026

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